



RAMAIAH INSTITUTE OF TECHNOLOGY, BANGALORE
(Autonomous Institute, Affiliated to VTU)
DEPARTMENT OF CIVIL ENGINEERING

Term: June to Sept 2023 **Course:** Introduction to Civil Engineering **Code:** ESC231

Max Marks: 30

Duration: 60 min

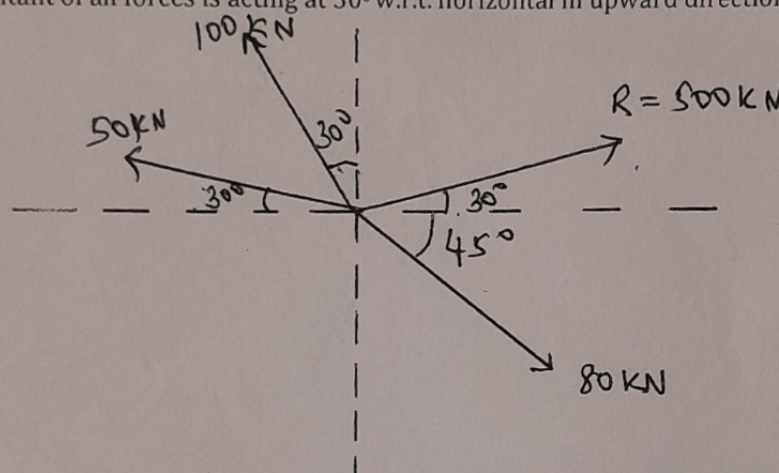
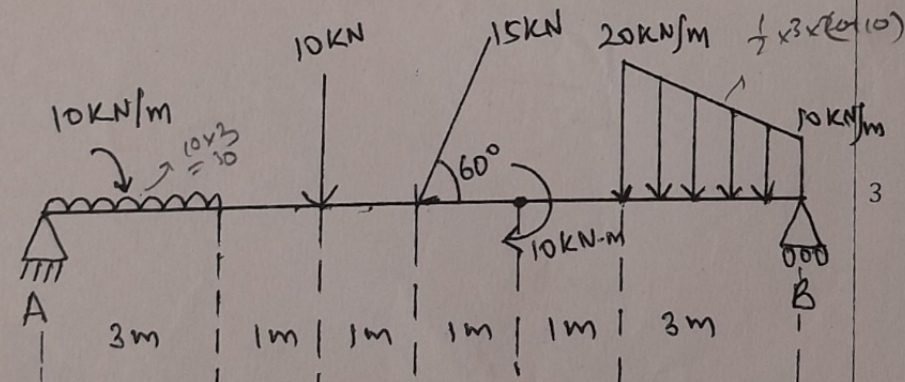
CIE: Test I

Sem: 2nd

Sec: --

Instructions to Candidates:

Answer any **TWO** full questions of the followings. Draw neat sketches wherever necessary

Sl#	Question	CO	BL	Marks
1 (a)	Discuss in detail scope of any two branches of Civil Engineering.	1	2	08
1 (b)	Discuss in brief the different types of force systems.	3	2	07
2(a)	Determine the magnitude, direction and position of the fourth force such that the resultant of all forces is acting at 30° w.r.t. horizontal in upward direction. 	3	3	08
2(b)	State and prove Varignon's theorem. Determine the support reactions for the system shown below. 	3	2	07
3(a)	Determine the support reactions for the system shown below.	3	3	08
3(b)	Discuss in detail the properties of good brick.	1	2	07

$$\sum F_H = 0$$

$$H_A - 15 \cos 60^\circ = 0$$

$$H_A = 7.5$$

$$\sum F_V = 0$$

$$V_A - 10 - 10 - 15 \sin 60^\circ - 45 + V_B = 0$$

$$V_A + V_B = 75.60$$

$$\sum M_A = 0$$

$$V_B(6) + H_A(0) + 10(1.5) + 10(3) + 15 \sin 60^\circ(5) + 10 + 45(7) - V_A = 0$$

$$V_B = 382.99$$

$$V_A = -307.99$$